STACKS, SHEAVES ON ALGEBRAIC STACKS: PROPOSITION-QC-COMPARE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5002-5008 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 5010-5128, the complete author proof (117 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_algebraic-stacks-sheaves.json (pb=4756, stmt 5002-5008, proof 5010-5128).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 09J1: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5128–5128. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition lemma-QC-compare: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4962–4968. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition lemma-compare: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1430–1452. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 08DW: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5128–5128. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition section-restriction-algebraic-spaces: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1421–1452. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GZ1: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5128–5128. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GZS: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5128–5128. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0H06: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5128–5128. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0H05: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 5128–5128. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let S be a scheme. Let $\mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids. Assume $\mathcal{X}$ is representable by an algebraic space X. Then $QC(\mathcal{X})$ is canonically equivalent to $D_{QCoh}(\mathcal{O}_X)$.*

Proof. Denote X_{affine} the category of affine schemes étale over X endowed with the chaotic topology and its structure sheaf $\mathcal{O}_X$, see Derived Categories of Spaces, Section 0H05. The functor $u : X_{étale} \to \mathcal{X}_{étale}$ of Lemma lemma-compare gives rise to a functor $X_{affine} \to \mathcal{X}_{affine}$. This is compatible with structure sheaves and produces a functor

$$G : QC(\mathcal{X}) = QC(\mathcal{X}_{affine}, \mathcal{O}) \longrightarrow QC(X_{affine}, \mathcal{O}_X)$$

See Cohomology on Sites, Lemma 0GZ1. By Derived Categories of Spaces, Lemma 0H06 the triangulated category $QC(X_{affine}, \mathcal{O}_X)$ is equivalent to $D_{QCoh}(\mathcal{O}_X)$. Hence it suffices to prove that G is an equivalence.

Consider the flat comparison morphisms $\epsilon_{\mathcal{X}} : \mathcal{X}_{affine,étale} \to \mathcal{X}_{affine}$ and $\epsilon_X : X_{affine,étale} \to X_{affine}$ of ringed sites. Lemma lemma-QC-compare and (the proof of) Derived Categories of Spaces, Lemma 0H06 show that the functors $\epsilon_{\mathcal{X}}^*$ and ϵ_X^* identify $QC(\mathcal{X}_{affine}, \mathcal{O})$ and $QC(X_{affine}, \mathcal{O}_X)$ with subcategories $Q_{\mathcal{X}} \subset D(\mathcal{X}_{affine,étale}, \mathcal{O})$ and $Q_X \subset D(X_{affine,étale}, \mathcal{O}_X)$. With these identifications the functor G in the first paragraph is induced by the functor

$$Li_X^* = R\pi_{X,*} : D(\mathcal{X}_{affine,étale}, \mathcal{O}) \longrightarrow D(X_{affine,étale}, \mathcal{O}_X)$$

where i_X and π_X are the morphisms from Lemma lemma-compare but with the étale sites replaced by the corresponding affine ones. The reader can show that this replacement is permissible either by reproving the lemma for the affine sites directly or by using the equivalences of topoi $Sh(\mathcal{X}_{affine,étale}) = Sh(\mathcal{X}_{étale})$ and $Sh(X_{affine,étale}) = Sh(X_{étale})$. The lemma also tells us Li_X^* has a left adjoint

$$L\pi_X^* : D(X_{affine,étale}, \mathcal{O}_X) \longrightarrow D(\mathcal{X}_{affine,étale}, \mathcal{O})$$

and moreover we have $Li_X^* \circ L\pi_X^* = \mathrm{id}$ since $\pi_X \circ i_X$ is the identity. Thus it suffices to show that (a) $L\pi_X^*$ sends Q_X into $Q_{\mathcal{X}}$ and (b) the kernel of Li_X^* is 0. See Derived Categories, Lemma 09J1.

Proof of (a). By Derived Categories of Spaces, Lemma 0H06 we have $Q_X = D_{QCoh}(X_{affine,étale}, \mathcal{O}_X)$. Let K be an object of Q_X. Let x be an object of $\mathcal{X}_{affine,étale}$ lying over the affine scheme $U = p(x)$. Denote $f : U \to X$ the morphism corresponding to x. Then we see that

$$R\Gamma(x, L\pi_X^* K) = R\Gamma(U, Lf^* K)$$

This follows from transitivity of pullbacks; see discussion in Section section-restriction-algebraic-spaces. Next, suppose that $x \to x'$ is a morphism of $\mathcal{X}_{affine,étale}$ lying over the morphism $h : U \to U'$ of affine schemes. As before denote $f : U \to X$ and $f' : U' \to X$ the morphisms corresponding to x and x' so that we have $f = f' \circ h$. Then

$$\begin{aligned} R\Gamma(x, L\pi_X^* K) &= R\Gamma(U, Lf^* K) \\ &= R\Gamma(U, Lh^* L(f')^* K) \\ &= R\Gamma(U', L(f')^* K) \otimes^{\mathbf{L}}_{\mathcal{O}(U')} \mathcal{O}(U) \\ &= R\Gamma(x', L\pi_X^* K) \otimes^{\mathbf{L}}_{\mathcal{O}(x')} \mathcal{O}(x) \end{aligned}$$

and hence we have (a) by the footnote in the statement of Cohomology on Sites, Lemma 0GZS. The third equality is Derived Categories of Schemes, Lemma 08DW.

Proof of (b). Let M be an object of $Q_{\mathcal{X}}$ such that $Li_X^* M = 0$. Let x' be an object of $\mathcal{X}_{affine,étale}$ lying over the affine scheme $U' = p(x')$ and assume that the corresponding morphism $f' : U' \to X$ is étale. Then $f' : U' \to X$ is an object of $X_{affine,étale}$ and the condition $Li_X^* M = 0$ implies that $M|_{U'_{étale}} = 0$. In particular, we see that $R\Gamma(x', M) = 0$. However, for an arbitrary object x of the site $\mathcal{X}_{affine,étale}$ there exists a covering $\{x_i \to x\}$ such that for each i there is a morphism $x_i \to x'_i$ with x'_i corresponding to an object of $X_{affine,étale}$. Now since M is in $Q_{\mathcal{X}}$ we have

$$R\Gamma(x_i, M) = R\Gamma(x'_i, M) \otimes^{\mathbf{L}}_{\mathcal{O}(x'_i)} \mathcal{O}(x_i) = 0$$

and we conclude that M is zero as desired. □